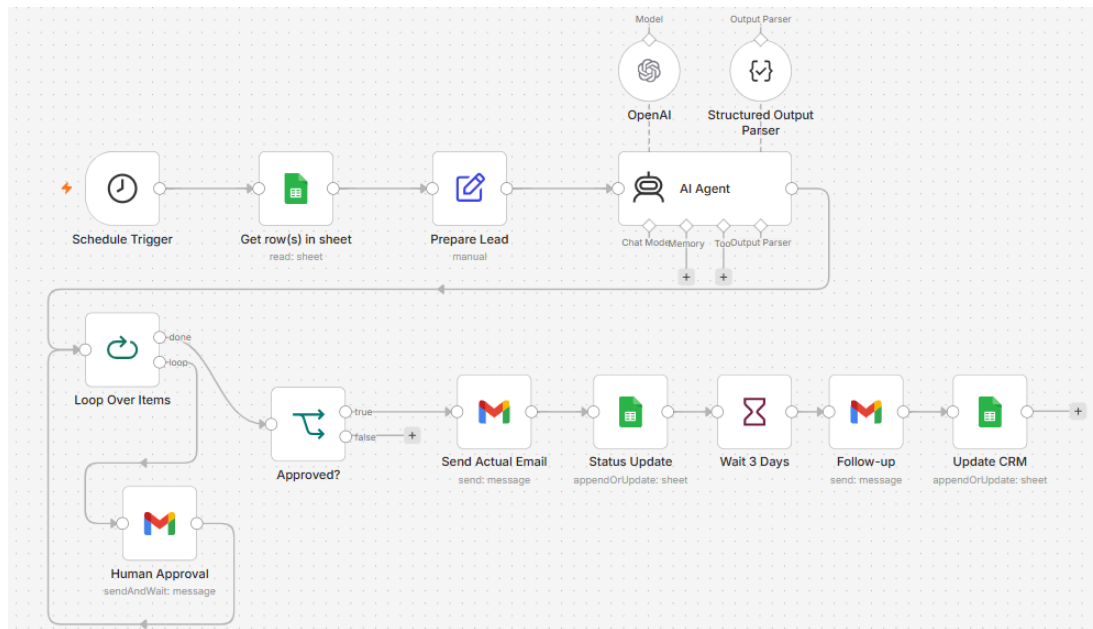


PORTFOLIO CASE STUDY

AI Sales Outreach & Follow-up Automation

n8n + Google Sheets + Gmail + AI



High-level workflow overview

01 — Project Overview

I built an automated B2B outreach workflow that finds leads from Google Sheets, generates personalized emails with AI, and sends them only after human approval.

After sending, the workflow automatically handles follow-ups and updates the lead status.

Lead Source	Personalize	Control	Send	Follow-up	Track
Google Sheets	AI Email	Human Approval	Gmail	Wait + Email	CRM Update

02 — The Problem

Manual outreach takes time, especially when teams need to:

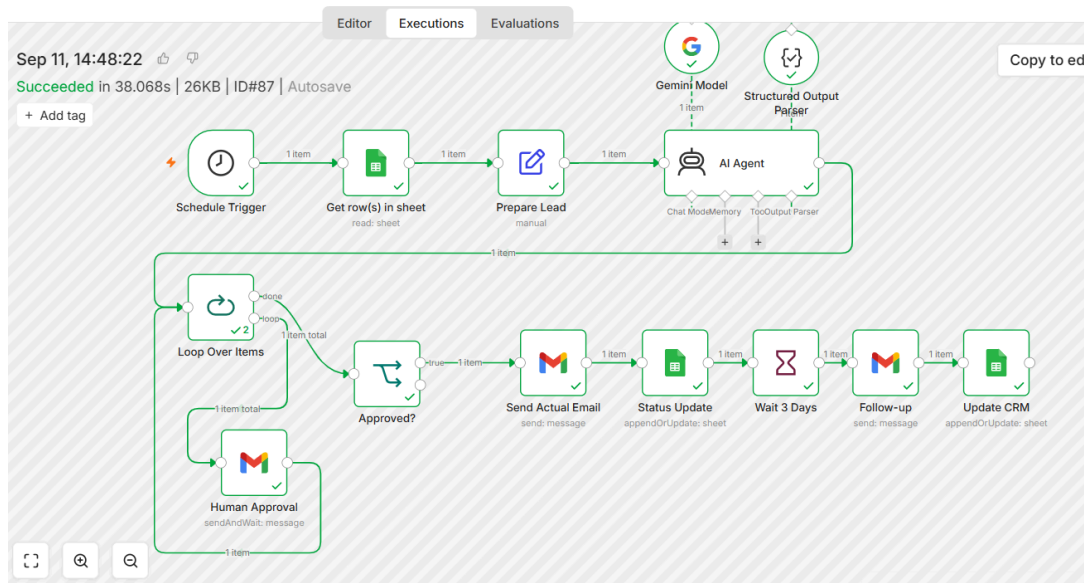
- Personalize every email
- Remember follow-ups
- Keep lead status updated
- Manage outreach at scale

Why this workflow matters

Without a proper process, follow-ups can be missed and tracking becomes repetitive.

03 — What I Built

A complete outreach workflow with four main stages.



End-to-end n8n workflow

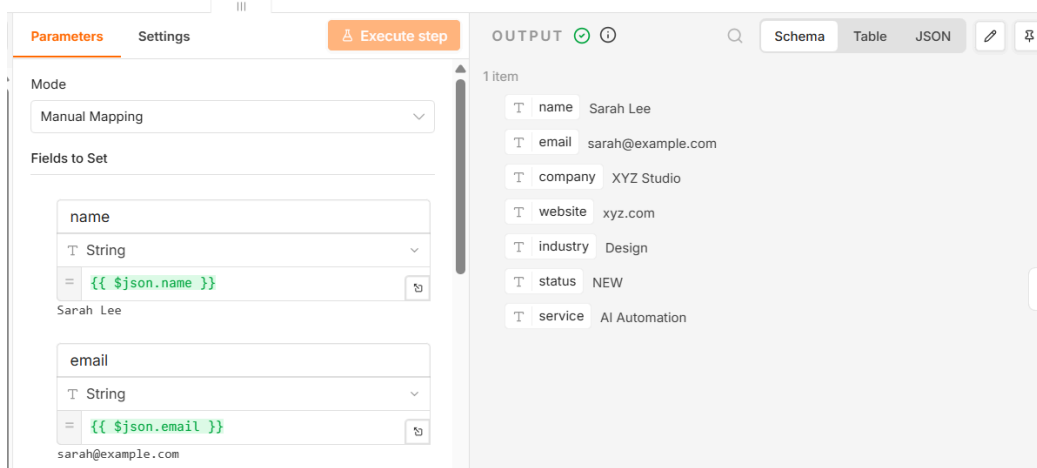
1	2	3	4
Lead Intake	AI Email	Human Approval	Send + Follow-up

Stage 1 — Lead Intake

The workflow pulls lead records on a schedule and prepares the data for AI processing.

Automation path

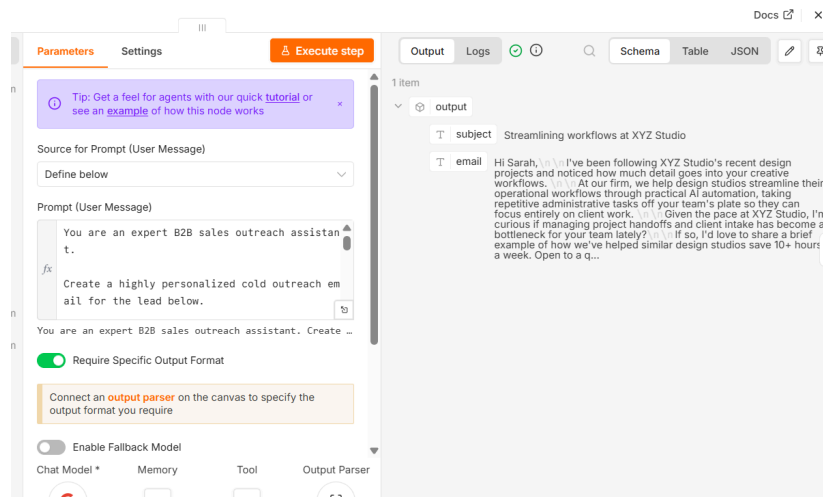
Schedule Trigger → Google Sheets → Prepare Lead



Lead data is prepared for the automation

Stage 2 — AI Email Generation

The AI creates a personalized outreach email based on the lead information.



AI agent generates the personalized email

The output contains:

Structured output

subject: "Personalized email subject" | email: "Personalized email body"

Using structured output keeps the email data ready for the next automation step.

Stage 3 — Human Approval

Before any outreach email is sent, a human reviews and approves it.

AI generated outreach email:

Company:
XYZ Studio

Lead:
Sarah Lee

Email:
sarah@example.com

Subject:
Streamlining creative workflows at XYZ Studio

Email:

Hi Sarah,

I've been following XYZ Studio's recent design projects and noticed how much detail goes into your creative workflows.

As a design studio, much of your team's time likely goes toward repetitive asset management, client intake, and administrative tasks rather than core creative work. We help design teams implement targeted AI automation to handle these routine processes.

Are you open to a brief chat next Tuesday to discuss how this could save your team a few hours each week?

Best,
Alex

Approve this email before sending?

Decline

Approve

Approval checkpoint before external communication

Draft	Review	Decision	Action
AI-generated	Human checks	Approved?	Send or stop

Control point

This adds a control point before external communication.

The design keeps automation responsible for repetitive work while a human controls the final outreach decision.

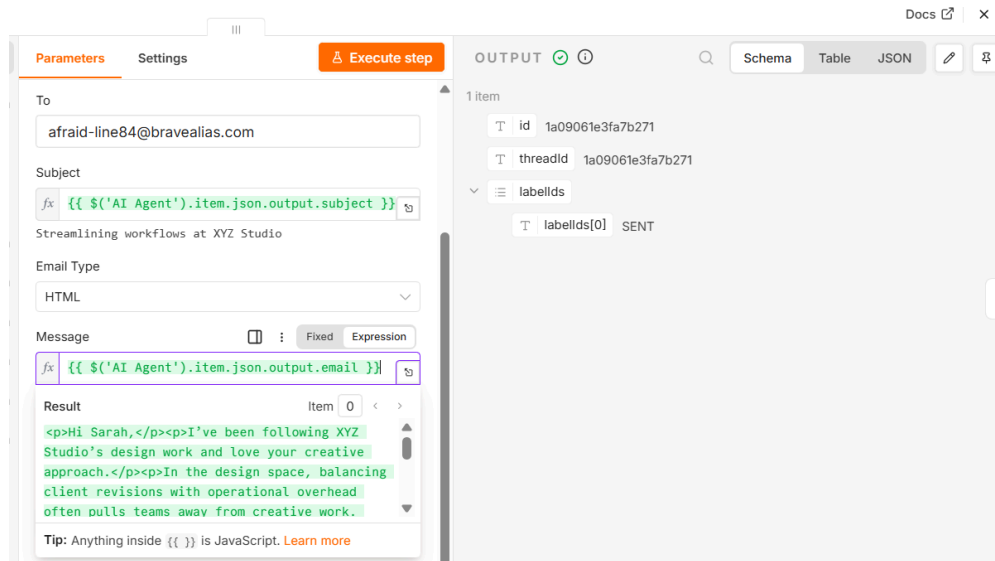
Stage 4 — Send, Follow-up & Tracking

After approval:

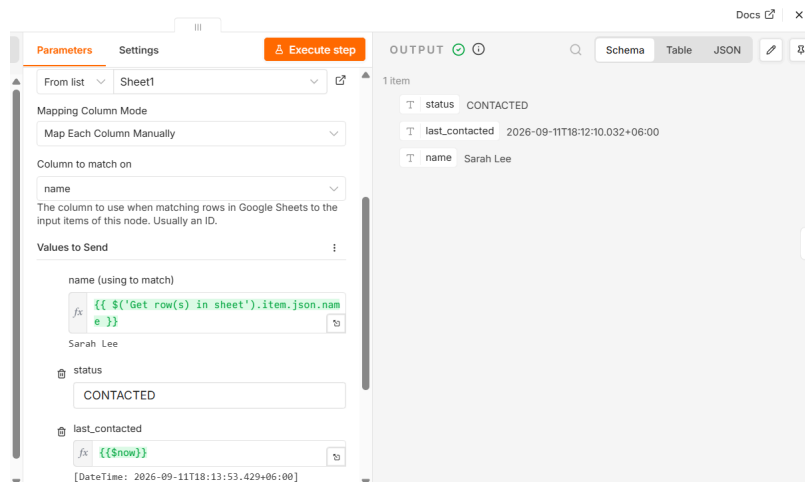
1. The email is sent.
2. The lead status is updated.
3. The workflow waits for the defined follow-up period.
4. A follow-up email is sent automatically.
5. The CRM/Sheet is updated again.

Automation path

Send Email → Update Status → Wait → Follow-up → Update CRM



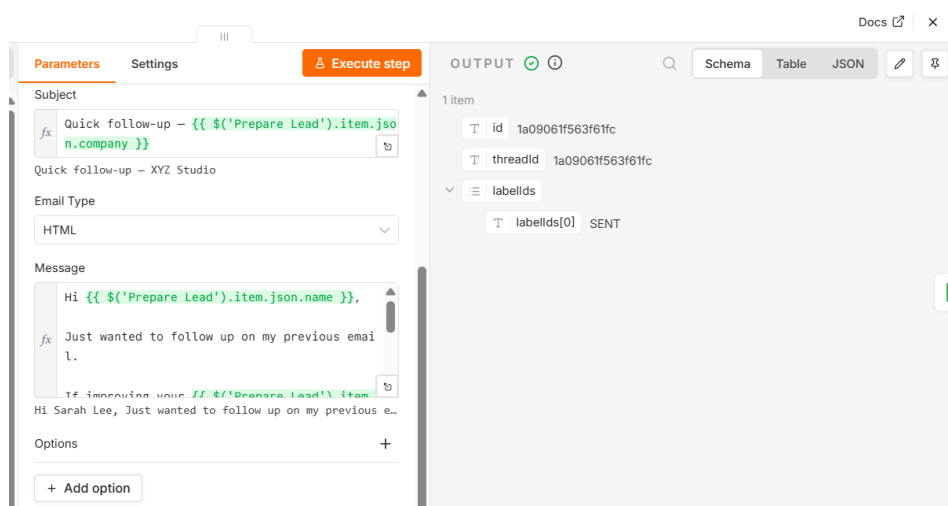
Send step and outbound email handling



Lead status update after sending

IMPLEMENTATION SCREENSHOTS

Follow-up Logic & CRM Updates



Follow-up email is generated from the prepared lead data

AI Sales Outreach CRM

File Edit View Insert Format Data Tools Extensions Help

Q

Menus

↶↷🖨🔊100%\$%\$%0.00123Default...-10+BI⌵A👉🏠📄📑🔍🗑🗑

Final lead record / CRM update

Resulting sequence

Send → Update Status → Wait → Follow-up → Update CRM

Together, these steps create a repeatable outreach loop rather than a one-off email send.

04 — Expected Business Outcome

This is a portfolio project, not a real client engagement, so no real performance metrics are claimed.

More consistent outreach A repeatable process reduces missed steps.	Less manual follow-up work Waiting and follow-up actions are automated.
Faster lead processing Lead records move into the outreach process automatically.	Better visibility into lead status The sheet / CRM is updated as the sequence progresses.
At scale Easier outreach at scale without turning every follow-up into a manual task.	

05 — Business Use Cases

B2B Sales Personalized prospecting and follow-ups.	Lead Generation Automated outreach after lead collection.
Marketing Agencies Client prospecting and campaign follow-up.	SaaS Sales Demo and trial outreach sequences.

06 — Key Takeaway

This system turns a spreadsheet of leads into an organized outreach process.

Prepare Lead data	Personalize AI email	Approve Human check	Send Gmail	Follow Up Wait + email	Track CRM / Sheet
-----------------------------	--------------------------------	-------------------------------	----------------------	----------------------------------	-----------------------------

The AI handles repetitive work, while human approval keeps the final outreach under control.

Tools & Technologies

Automation n8n Schedule Trigger • Switch • Wait • Human Approval	Data + Email Google Sheets • Gmail
AI Google Gemini	Reliability Structured Output Parser